Khushi Kaushik

Computer Science (AI) Undergraduate | Machine Learning | Computer Vision | Python | MATLAB
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Professional Summary

Computer Science undergraduate specializing in Artificial Intelligence at Banasthali Vidyapith. Experienced in Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), and Explainable AI through academic and research projects. Developed DeepX, a multimodal deepfake-detection framework selected for the IHCI Pre-Conference (2025), and a Retrieval-Augmented Generation (RAG) document QA application. Holds five MathWorks certifications in MATLAB, machine learning, image processing, and computer vision.

Education

Banasthali Vidyapith, Jaipur *May 2027 (Expected)*
B.Tech in Computer Science with Artificial Intelligence *CGPA: 8.03 / 10.0*
Coursework: Image Processing and Computer Vision (IPCV) Lab, Geoinformatics

Delhi Public School, Ranipur *Class XII: 76% (2023) | Class X: 95.2% (2021)*

Technical Skills

- **Programming Languages:** Python, MATLAB, Java, C++, C, JavaScript, SQL
- **AI/ML and Computer Vision:** Machine Learning, Deep Learning, TensorFlow, OpenCV, Image Processing, Natural Language Processing (NLP), Explainable AI, Grad-CAM, LIME
- **RAG and LLM Technologies:** Retrieval-Augmented Generation (RAG), Sentence Transformers, FAISS, Vector Embeddings, Semantic Search, Groq API
- **Application Development:** Flask, Django, React.js, Streamlit, REST APIs
- **Databases and Developer Tools:** MySQL, MongoDB, Firebase, Git, GitHub, GitHub Actions, CI/CD, Linux, LaTeX
- **Core Computer Science:** Data Structures and Algorithms, Object-Oriented Programming, Software Testing, Data Analysis

Projects

DeepX — Explainable Multimodal Deepfake Detection Framework *Sep 2025 – Feb 2026*
Research with Dr. Aditi Paul | Python, TensorFlow, OpenCV, Librosa, Flask, Django, MySQL

- Developed a multimodal deepfake-detection framework for image, audio, and video using CNN-based modelling, MFCC audio features, and video-frame embeddings.
- Integrated Grad-CAM heatmaps and attention mechanisms to support visual interpretation and audio-anomaly analysis.
- Applied MATLAB, image processing, and computer vision concepts to support model development and research analysis.
- Documented the methodology with Dr. Aditi Paul; the research was selected for the IHCI Pre-Conference (2025) and featured in the college research magazine.

RAG Evaluation and Hybrid Retrieval System *Aug 2026 – Oct 2026*
Python, Sentence Transformers, FAISS, BM25, Cross-Encoder Reranking, Groq API, Streamlit, pytest

- Designed a hybrid retrieval approach combining dense vector search with BM25 keyword search to retrieve semantic content and exact technical terms from PDF documents.
- Planned evaluation using Recall@k, Mean Reciprocal Rank (MRR), answer relevance, and citation correctness on a curated question-and-source dataset.
- Designed source-grounded answer generation with page-level citations and evidence-sufficiency checks to help identify unsupported questions.
- Planned automated regression tests for document ingestion, retrieval, citation mapping, and error handling using pytest and GitHub Actions.

Tech Stack: Python, FAISS, BM25, Sentence Transformers, Groq API, Streamlit, pytest, GitHub Actions

Certifications

MathWorks (2025):

- MATLAB Onramp
- Machine Learning Onramp
- Image Processing Onramp
- Computer Vision Onramp
- Color Spaces and Image Segmentation

Achievements & Activities

- Presented the DeepX research at the IHCI Pre-Conference, Banasthali Vidyapith (2025).
- Active IEEE member; participated in the IEEE Computer Society Innovation and Leadership in AI Workshop (2025).